

VPN and Remote Access Security — Guided Notes

Unit 4.2.2 | CompTIA Network+ N10-009 Obj. 4.3 | N10-008 4.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

IPsec uses two protocols: _____ (AH) provides authentication and integrity but no encryption, and _____ (ESP) provides encryption, authentication, and integrity. Which is used in modern VPN deployments and why?

Question 2

IPsec tunnel mode encapsulates the _____ original IP packet inside a new IP packet. Transport mode protects only the _____, leaving the original IP header visible. Which mode is standard for site-to-site VPNs?

Question 3

IKE Phase 1 establishes a _____ channel used for negotiation, authenticating peers via _____ or certificates. IKE Phase 2 uses this channel to negotiate the _____ that will protect actual data traffic.

Question 4

SSL/TLS VPNs operate over TCP port _____, which makes them traverse most firewalls and NAT devices easily. IPsec VPNs can struggle with NAT because NAT modifies _____, breaking IPsec's integrity verification.

Question 5

In a full-tunnel VPN, _____ client traffic flows through the VPN. In a split-tunnel VPN, only traffic destined for _____ routes through the tunnel. What is the security tradeoff of split tunnel?

Question 6

Multi-factor authentication (MFA) for VPN access requires something you _____ plus something you _____. Why is username and password alone insufficient for VPN access?

Question 7

Endpoint posture checks evaluate the _____ device before granting full VPN access. List two conditions a posture check might require before allowing connection.

Question 8 — Real World Application

REAL WORLD: A former contractor's VPN account was never deactivated. Three months after their contract ended, unauthorized access to internal project files is discovered through that account. Identify two process failures that allowed this and state the control that prevents each.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-2-2-vpn-and-remote-access-security/4-2-2-vpn-and-remote-access-security-answer-key.pdf
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